

**We Have Never Been Intelligent:
The Cognitive Constitution and the Myth of the Individual
Mind**

PIATRA . INSTITUTE

June 2026

Abstract

Human cognitive capacities are real and unequal, yet the performances offered as evidence of intelligence, in mathematics, physics, law, engineering, administration, and artificial intelligence, are seldom produced by an isolated brain. They are realized by assemblages of trained bodies, notations, instruments, collaborators, procedures, and inherited institutions, and are then attributed to a single named person. We analyse this attribution with Bruno Latour's account of modernity, in which a public doctrine separating nature from society coexists with a practice that multiplies nature-society hybrids. An analogous Cognitive Constitution governs intelligence. Institutions depend operationally on cognitive hybrids and credit their outputs to bounded individuals through attributional compression, which books collaborative production to a genius, an educational infrastructure to an examination score, a software ecosystem to a programmer's talent, and a corpus and a data centre to a model. Separating four referents of the folk concept (capacity, performance, realization, and historical possibility) shows psychometrics, distributed cognition, and cultural evolution to be accounts of different levels. The argument requires only that the realizing system for many advanced tasks exceeds the individual and that purification proceeds through observable attribution practices; it sets aside the thesis that artifacts are constitutive parts of the mind. A contested proposal, that the capacity to learn is a generic property of mathematical structure found by selection, would extend the compression to human biology. Attributional compression persists because it is the accounting system of merit: examinations, prizes, wages, and liability each require a single body to credit. In its defensible form the thesis is that humans have never been intelligent alone.

1. Introduction

A physicist writes a line of symbols on a blackboard that resolves a problem, and the room agrees that it has watched intelligence at work. The room also contains three centuries of notation, without which the line could be neither written nor read; the board, a surface on which the hand holds intermediate steps that working memory cannot; two decades of training that installed the moves the hand now makes without deliberation; the literature the result answers and the community that will check it; the instruments that will test it; and the institution that pays for all of this and will, if the result holds, attach the physicist's name to the achievement and to no one else's. Everyone present can see these components, and everyone present credits the person.

That cognition is distributed is by now a settled finding. The phenomenon under analysis is that the individual continues to be credited in full view of the distribution, and that the crediting is an operation on which institutions depend. The gap between what produces a performance and what receives credit for it is stable, structured, and productive, and it can be described.

Read literally, the title is false. Humans learn, abstract, plan, and reason, and they differ in how well they do so, reliably enough that the differences predict outcomes (Gould, 1981; Carson, 2007). The defensible claim is narrower. Humans have never been intelligent in the purified sense the word usually carries, in which intelligence is an inner property of a bounded individual who then applies

it to the world. The sciences have not found intelligence in that sense inside people; institutions assign it to people, and the assignment is the subject of the analysis. The precise form of the thesis adds one word: we have never been intelligent alone.

The method is borrowed from Bruno Latour, who made a structurally parallel argument about modernity. The analysis extends it in two directions Latour did not take: it identifies the counter-part of his purification for the case of mind, and it explains why that purification, unlike a confusion, persists after it has been pointed out.

2. Latour's Modern Constitution

Latour's claim in *We Have Never Been Modern* is that modernity never matched its own description (Latour, 1993). The modern settlement, which he calls the Constitution, rests on a purification. It sorts the world into two zones, an objective nature discovered by science and a subjective society negotiated by politics, and it polices the border between them. The work of the moderns is to keep the zones apart, to insist that a fact is a fact, a value is a value, and nothing belongs to both.

The civilization that purifies most strenuously also manufactures, faster than any before it, objects that belong to both zones at once. The hole in the ozone layer is simultaneously a fact of atmospheric chemistry, a product of industrial policy, and a problem for international law, and no procedure divides it into a clean fact and a clean value. Latour calls such objects hybrids and argues that modernity produces them in quantity because it refuses to admit that they exist. The Constitution functions because it forbids the hybrid: by guaranteeing that nature and society are separate, it frees its practitioners to mix them without limit, since officially no mixing occurs.

The laboratory shows this most clearly, and the early laboratory studies are where Latour first described it (Latour and Woolgar, 1986; Latour, 1987). A laboratory builds a hybrid, an entity that is at once a natural phenomenon and a social construction, and then performs the purification that allows the phenomenon to leave as nature with no trace of its making. What leaves is the inscription, the graph, number, or printed result, stripped of the apparatus, labor, and negotiation that produced it, and the inscription travels and counts as the fact while the apparatus stays behind. Ethnographers of other epistemic cultures describe the same movement from the inside: the machinery by which a result is made and the disappearance of that machinery from the result (Knorr Cetina, 1999; Galison, 1997). Purification is the name for that disappearance.

Two features of Latour's argument transfer to intelligence and carry most of the analysis below. The first is the double structure, a public doctrine of separation running over a private practice of mixture. The second is directionality, an operation that strips the network from the product so that the product can be credited to one of the two pure zones. The third feature, symmetry, does not transfer. Latour wanted nature and society treated alike. The case of intelligence is asymmetric, because the purified pole is a person, and institutions have strong and independent reasons to require persons.

3. The Cognitive Constitution

Applied to intelligence, the same analysis yields a parallel settlement, here called the Cognitive Constitution. Its public doctrine holds that intelligence is an inner property of the individual mind, a quantity a person possesses and brings to tasks, visible most clearly when the person works alone and unaided. Its private practice is that almost nothing identified as intelligence is produced that way. Advanced performances are made by assemblages, couplings of trained brains with notations, diagrams, instruments, collaborators, divisions of labor, and inherited institutional procedure, and the daily work of the Constitution is to take the output of an assemblage and book it to a person.

The hybrids in this case are cognitive. The extended-mind tradition named the smallest of them, the coupling of a brain with an external resource used as part of its thinking, and argued that under suitable conditions the resource is a component of cognition in addition to an aid to it (Clark and Chalmers, 1998; Clark, 2008). The distributed-cognition tradition named the largest, cognitive systems spanning many people and artifacts that compute what no member computes alone (Hutchins, 1995; Hollan, Hutchins and Kirsh, 2000). Between these sit the embodied and enactive accounts, for which thinking is an activity of the whole acting body in a structured environment (Varela, Thompson and Rosch, 1991; Noë, 2004; Gallagher, 2005), and the material-engagement account, for which the clay, the wheel, and the abacus are sites where thought happens (Malafouris, 2013). The term cognitive assemblage predates this analysis: Hayles uses it for the couplings of human and technical cognizers through which information moves and decisions are made (Hayles, 2017).

Together these traditions establish that the hybrid is real and pervasive. None of them explains the purification. If cognition is so plainly spread across the assemblage, the individual mind should not remain the unit of every examination, prize, wage, and obituary, yet it does. The explanation lies in an operation that institutions perform on cognition, and the remaining sections describe that operation. Table 1 sets out the correspondence with Latour's scheme.

Latour's Modern Constitution	The Cognitive Constitution
Public doctrine: nature and society are separate	Public doctrine: intelligence is an inner individual property
Private practice: laboratories manufacture nature-society hybrids	Private practice: institutions run cognition on brain-tool-group-institution hybrids
Purification strips the network from the fact	Attributional compression strips the assemblage from the performance
The inscription travels; the apparatus disappears	The score, the byline, the credential travels; the assemblage disappears
Works because forbidding the hybrid licenses making it	Works because crediting the individual licenses relying on the system
Product credited to pure nature	Product credited to the bounded person

4. Capacity, Performance, Realization, and Historical Possibility

The folk concept of intelligence fuses several referents, and many long disputes about intelligence are disputes between parties who hold different members of the fusion. Separating them costs nothing and supports the rest of the argument. Four referents travel under the one word.

The first is capacity, what a person can reliably learn and contribute across situations. Capacity is the proper object of psychometrics; it varies between people, the variation is partly heritable, and it predicts outcomes. The thesis does not require denying any of this, and the denial would be false (Gould, 1981; Carson, 2007). The second is performance, what is accomplished on a particular task on a particular day. The third is realization, the system whose operations produce that performance, which may be a brain alone but for advanced tasks is usually a brain coupled to much else. The fourth is historical possibility, the inherited infrastructure of notation, technique, and institution that makes the task thinkable; without it the performance is unavailable altogether.

These are questions about different levels, and the sciences that address them do not conflict. Psychometrics measures capacity and finds it unequal. Distributed cognition describes realization and finds it supra-individual. Cultural evolution accounts for historical possibility and finds it cumulative, the product of a ratchet no individual turns alone (Boyd and Richerson, 1985; Henrich, 2016). A child can become a physicist and a squirrel cannot. The child's brain does not contain physics; the child can be enrolled into the cumulative culture that does, through the shared intentionality and high-fidelity social learning that are the specialism of the human animal (Tomasello, 1999; Tomasello et al., 2005; Heyes, 2018; Donald, 1991). The capacity is real and located in the child. The performance, when it comes, will be realized by an assemblage and made possible by three centuries of inherited apparatus. The folk concept assigns all four to the child and calls the sum the child's intelligence.

5. Extension, Scaffolding, and Attributional Compression

The literature that established the cognitive hybrid spent two decades on a metaphysical dispute. Clark and Chalmers proposed that Otto's notebook, which an amnesiac patient consults as another person consults memory, is part of Otto's cognitive system, and the debate that followed asked whether the notebook is literally part of the mind or a tool the mind uses (Clark and Chalmers, 1998). Critics replied that causal dependence does not amount to constitution, that a thing can matter greatly to cognition without being cognitive, and that the mark of the cognitive does not extend to ink (Adams and Aizawa, 2008; Rupert, 2004). Sterelny proposed a middle position: most of what the extended-mind theorists describe is better called scaffolding, an environment built over a life and a history to support and shape a mind that remains, for bookkeeping purposes, inside the skin (Sterelny, 2010; Sterelny, 2012).

The metaphysical question is genuine, but the thesis does not depend on it and should not be rested on it. Grant the critics their full case: the notebook is a tool, constitution is a strong word that the cases seldom earn, the person at the board remains the cognitive agent, and the board remains a board. The phenomenon is unaffected, because it concerns attribution. The performance is realized

by the whole coupled system and credited to one component of it. Whether the board is part of the physicist's mind is a question for philosophers. That the physics is produced by the physicist together with board, training, and community and then awarded to the physicist is a question for those who confer awards, and they have already settled it.

The argument therefore relies on the modest claim, which the critics also grant. For many advanced tasks the realizing system demonstrably exceeds the individual, in the sense that the same person deprived of the notation, the instruments, or the team fails at the task outright (Hutchins, 1995; Kirsh and Maglio, 1994). The modest claim leaves a question that the distribution literature did not take up. Given that the distribution is visible to everyone, what function does continued crediting of the individual serve, and by what mechanism is the crediting performed? We call the mechanism *attributional compression*. It takes a performance distributed across an assemblage and returns a property of a person, as the laboratory takes a hybrid and returns a fact. It is lossy by design, and the loss is functional: a score, a byline, a grade, a citation, or a salary is a single number or a single name, and a single name is what downstream institutions require. Compression is the intended output of the system.

6. Six Cases of Attributional Compression

The same structure appears wherever cognition is impressive enough to attract credit, and that correlation is itself diagnostic. Six instances follow.

Quantum mechanics is understood by no unaided mind, including the minds that use it best. It is realized by a coupling of formalism, the standardized notation that carries the inference, with laboratory apparatus that produces the phenomena, pedagogy that installs the moves, textbooks that fix the consensus, and peer review that certifies it. Without the formalism the most able physicist could not reconstruct the subject from intuition, because the intuition derives from the formalism. What appears as understanding a theory is competent participation in an assemblage that holds the theory.

Mathematics is the clearest case because its scaffold is visible on the page. Its history is in large part a history of notation, and the notation is the machinery of mathematical thought. The Greek diagram is a device that makes deduction possible (Netz, 1999), and algebraic and symbolic writing are technologies for thinking what would otherwise be unthinkable, so that the mathematician is, in a strong sense, something the symbols do (Rotman, 2000). A proof is a social and material procedure before it is a private insight, and the blackboard is where the result is computed.

Navigation is the case Hutchins observed directly. A naval vessel fixes its position through a process spread across a team, a set of instruments, a chart, and a fixed sequence of procedure; the computation is a property of the bridge, and the system remembers, computes, and corrects in ways no individual sailor does (Hutchins, 1995). The navigator is a node with a job, and a well-run bridge can absorb a weak node and continue to navigate, which is the operational signature of cognition located somewhere other than where the credit goes.

The laboratory, Latour's case, is the template for the others. Knowledge leaves it as inscription, with apparatus, labor, and calibration removed, so that the result reads as a fact about nature and the achievement of the network is lost from view (Latour and Woolgar, 1986; Knorr Cetina, 1999). The trained perception that allows a researcher to see the result at all is itself an institutional product, drilled in over years until it feels like vision (Daston and Galison, 2007).

Software is the contemporary and most literal case. A working program is realized by a programmer coupled to a codebase written mostly by others, a compiler that encodes rules no programmer remembers, version control that serves as the project's external memory, test suites that do the checking, documentation, and a global archive of solved problems consulted continuously. The programmer is the interface to a technical-symbolic system far larger than any individual, and the talent the industry pays for consists largely of fluency at coupling. Without the archive and the compiler the talent has nothing to operate on.

Artificial intelligence is the case that has made the others legible. The intelligence of a model is a corpus on the scale of a civilization's writing, a data centre's worth of compute, an architecture, a benchmark culture, and a large quantity of poorly paid human feedback, compressed into a noun and a brand. This is the compression the Cognitive Constitution performs on people, carried out quickly and recently enough that the assemblage remains visible before the noun closes over it. Artificial intelligence did not introduce the problem of crediting a system's output to an agent. It exhibits a problem that human institutions had already settled for people by ceasing to notice it.

7. Attribution as an Accounting System

A mistake as durable as the individual mind would be the most persistent error in the human sciences, and durability of that order usually indicates that something is being accomplished. The Cognitive Constitution functions as an accounting system, and what it accounts for is merit.

The individuated number is needed at identifiable points. A school must decide who advances, and it cannot advance an assemblage, so it builds an instrument, the closed-book examination, designed to remove the scaffold and read off the remainder, and it calls that residue the student's intelligence. The history of the instrument is a history of deliberate purification: the construction of a unit of mind that institutions could test, rank, and compare, carried out in the same decades that built the meritocratic state and for the same purpose (Danziger, 1990; Carson, 2007; Gould, 1981). A science must decide whom to reward, and it cannot award a prize to a network, so it operates a credit economy that resolves collective production into named authorship, a convention whose workings the social history of science has documented (Shapin, 1994). A firm must set wages, a court must assign responsibility, and a state must identify the citizen who votes, is punished, and is owed. Each requires the output of a system to attach to a body, because bodies are what can be paid, promoted, blamed, and enfranchised. An assemblage cannot hold a credential or stand trial.

This is where the asymmetry noted in Section 2 matters. Latour could ask for nature and society to be treated symmetrically because neither needed to be a person. Intelligence cannot be symmetrized in that way, because its purified pole is the individual, and the individual is the in-

dispensable unit of every institution that distributes anything. Demonstrating that cognition is distributed does not dissolve the individual mind, as demonstrating that value is social does not dissolve money. The demonstration is correct and leaves the institution unchanged, because the institution assigns credit and needs a single name to do so.

The folk concept of intelligence therefore performs a function, and the function is settlement. It determines, before any inquiry into where the thinking occurred, that the thinking will be charged to a person, because the institutions that consume the answer can bill only a person. Attributional compression is the operation that produces the bill. The individual mind is an output of that operation, and this explains its survival through every finding that ought to embarrass it.

8. The Biological Floor and the Free-Lunch Hypothesis

The regress of credit appears to end at biology. Credit can be moved from the individual to the assemblage and from the assemblage to the cultural ratchet, and it still seems to reach solid ground at the species, since the assemblage is a human artifact, the ratchet a human achievement, and the capacity to build and accumulate an endowment that selection took a very long time to install in one animal and not in others. Every relocation so far stays inside the human story. On a deflationary reading, intelligence would bottom out in a fact about evolved brains, the fact on which Sections 4 and 9 rely. A recent argument proposes that there is no such floor.

The argument comes from Michael Levin and collaborators. It holds that the capacity on which the analysis has rested, the ability to learn and to organize behavior toward a problem, is found by natural selection and not constructed by it. In Levin's framing the competencies of living things ingress from a pre-existing space of mathematical structure instead of being assembled anew in each lineage, so that bodies are interfaces onto patterns that are discovered (Levin, 2025a; Levin, 2022). The strongest version, presented in a recent lecture over a slide credited to Federico Pigozzi, is that the intelligence ratchet is a gift from mathematics, as opposed to physics or biology: random, unselected networks already lie close to optimal for learning and for causal organization, so that selection encounters a free lunch and exploits it (Levin, 2025b).

The proposal has published support and remains unsettled. On the published side, Pigozzi, Goldstein, and Levin (2025) report that associative conditioning raises integrative causal emergence in gene-regulatory network models, and they propose a feedback loop, which they name an intelligence ratchet, in which emergence improves learning and learning in turn raises emergence; Pigozzi and Levin (2026) find that the same measure of causal emergence predicts the final reward of reinforcement-learning agents. Beneath this lies the earlier formal result that a coarse-grained macro description can carry more causal power than the micro substrate it summarizes (Hoel, Albantakis, and Tononi, 2013; Hoel, 2017), extended to networks by Klein and Hoel (2020), and several independent lines of work treating learning as a generic property of dynamical systems in addition to a product of selection (Watson and Szathmáry, 2016; Vanchurin, Wolf, Katsnelson, and Koonin, 2022), some of which place emergent causal structure before the first replicators (Walker and Davies, 2013). On the unsettled side, the strongest claims, the near-optimality of random networks and the assertion that mathematics and not biology drives the ratchet, have so far been

presented in talks and work in progress, and the metaphysics of a literal realm from which forms ingress is contested and probably unfalsifiable.

The analysis does not need that metaphysics and declines it for the reason it declined the claim that the notebook is part of the mind. Suppose there is no realm from which forms come, and that a gift from mathematics is a vivid way of reporting that random networks are unexpectedly learnable. The deflationary content survives, because it concerns the source of the capacity. On this reading the capacity comes from no particular place: it is a generic affordance of structured matter that selection encountered and that the human lineage happens to couple to better than most.

On that reading the argument of the preceding sections extends to a further scale. Attributional compression credits the individual for the work of the assemblage. The regress can be halted at the species by crediting human biology with the capacity to learn. If learning is a free lunch, however, crediting the evolved brain with intelligence is the same compression at one further remove: a generic affordance, available in principle to a random network, booked to an animal as the assemblage is booked to a genius and the corpus to a model. The names change and the operation is the same. The individual did not author the performance, and on this account the species did not author the capacity. Credit is assigned at every level and authorship located at none.

This conflicts with the biological fact used in Sections 4 and 9, that one animal can be enrolled into cumulative culture at high fidelity, which the free-lunch hypothesis appears to dissolve. Two considerations resolve the conflict. First, the levels do not collide, since a generic capacity to learn does not make every interface onto it equal, and the human animal can be the body that couples best to the ratchet even if no animal invented the ratchet, which is Levin's own interface picture. Second, the thesis concerns attribution and does not require a decision about the ultimate source of cognition. Whether the source is the cortex, the culture, or mathematics, the institution still needs one name to charge and charges the nearest body, and the further the actual source lies from that body, the more clearly the name on the result reads as a decision. If the free-lunch hypothesis holds, it moves the individual mind further still from anything it could be a fact about.

A caution follows from the same standard. A gift from mathematics can be a precise claim about the learnability of random networks, or it can be a re-enchantment that moves genius out of the skull and into the cosmos. The second reading is a version of the intelligence-mysticism discussed in Section 9. Only the first reading, the precise claim, is used here.

9. Scope and Limits

Relocating intelligence leaves four things in place.

It does not deny biology, although Section 8 qualifies the standing of biology. At one level the thesis relies on a specific, located, and unequal biological fact: the human animal can be enrolled into cumulative culture at high fidelity through capacities for imitation, shared attention, and teaching that other animals lack or hold weakly (Tomasello, 1999; Heyes, 2018; Henrich, 2016). If the free-lunch hypothesis is correct, that enrollability is the quality of one interface onto a capacity older and more generic than any lineage, and not the origin of the capacity to learn. On either account the

fact is real, resides in individual organisms, and varies between them, and the argument relocates the production of advanced performance without emptying the person who is enrolled. A brain is one part of the assemblage, and the part without which it cannot run; an assemblage monism that dissolved the brain into the network would be as one-sided as the individualism it replaced.

It does not abolish individual differences; it specifies what they are differences in. The variance that tests capture is real and residual by construction, what remains after the testing room has held the scaffold constant, removed the tools, excluded collaborators, and equalized the notation in which everyone was already trained. The test purifies deliberately and then reports the purified residue as the person's intelligence, the Constitution operating in miniature (Gould, 1981; Danziger, 1990). Measuring capacity in this way is legitimate. Treating the measurement as the location of intelligence is the error identified here.

It does not abolish agency, genius, or contribution; it redescribes them as coupling. The navigator, the mathematician, and the programmer each contribute something indispensable, and removing any of them collapses the performance as surely as removing the notation. What changes is the grammar: each contribution is a contribution to an assemblage, the skilled coupling of a trained body to a system that exceeds it, and skilled coupling is real, unequal, and creditable. Tacit knowledge, the part of expertise that cannot be written down and must be acquired through participation, is the skill of coupling, and it belongs to the person although it cannot be stated (Polanyi, 1966; Collins, 2010; Lave and Wenger, 1991; Dreyfus, 1992).

It does not erase the difference between human and artificial intelligence, although it explains their apparent resemblance. The resemblance lies in the attribution, the output of a system booked to an agent, and not in the systems themselves. The human is scaffolded by a culture into which it is born and trained by other bodies; the model is scaffolded by a corpus on which it is trained and tuned by human labor. The parity is in the bookkeeping. Recognizing the shared bookkeeping counters the mysticism on both sides: the genius-mysticism applied to people and the intelligence-mysticism now being applied to models.

10. Conclusion

Advanced cognitive performances are realized by assemblages and credited to individuals, and the crediting is performed by attributional compression, the operation through which examinations, bylines, credentials, wages, and liability attach the output of a system to a single body. The folk concept of intelligence fuses capacity, performance, realization, and historical possibility; separating them reconciles psychometrics, distributed cognition, and cultural evolution, and it shows why demonstrations of distribution leave the individual mind in place. The current disputes over artificial intelligence, over who built it, who owns it, and whose labor is compressed inside it, are also disputes over how intelligence has been attributed to individuals since institutions began attaching examinations, prizes, and wages to them. Artificial intelligence has reopened a settlement about credit, and the attribution of cognition to persons was always a part of that settlement.

References

- Adams, F., and Aizawa, K. (2008). *The Bounds of Cognition*. Blackwell.
- Boyd, R., and Richerson, P. J. (1985). *Culture and the Evolutionary Process*. University of Chicago Press.
- Carson, J. (2007). *The Measure of Merit: Talents, Intelligence, and Inequality in the French and American Republics, 1750–1940*. Princeton University Press.
- Clark, A. (2008). *Supersizing the Mind: Embodiment, Action, and Cognitive Extension*. Oxford University Press.
- Clark, A., and Chalmers, D. (1998). The extended mind. *Analysis*, 58(1), 7–19.
- Collins, H. (2010). *Tacit and Explicit Knowledge*. University of Chicago Press.
- Danziger, K. (1990). *Constructing the Subject: Historical Origins of Psychological Research*. Cambridge University Press.
- Daston, L., and Galison, P. (2007). *Objectivity*. Zone Books.
- Donald, M. (1991). *Origins of the Modern Mind: Three Stages in the Evolution of Culture and Cognition*. Harvard University Press.
- Dreyfus, H. L. (1992). *What Computers Still Can't Do: A Critique of Artificial Reason*. MIT Press.
- Gallagher, S. (2005). *How the Body Shapes the Mind*. Oxford University Press.
- Galison, P. (1997). *Image and Logic: A Material Culture of Microphysics*. University of Chicago Press.
- Gould, S. J. (1981). *The Mismeasure of Man*. W. W. Norton.
- Hayles, N. K. (2017). *Unthought: The Power of the Cognitive Nonconscious*. University of Chicago Press.
- Henrich, J. (2016). *The Secret of Our Success: How Culture Is Driving Human Evolution, Domesticating Our Species, and Making Us Smarter*. Princeton University Press.
- Heyes, C. (2018). *Cognitive Gadgets: The Cultural Evolution of Thinking*. Harvard University Press.
- Hoel, E. P. (2017). When the map is better than the territory. *Entropy*, 19(5), 188.
- Hoel, E. P., Albantakis, L., and Tononi, G. (2013). Quantifying causal emergence shows that macro can beat micro. *Proceedings of the National Academy of Sciences*, 110(49), 19790–19795.
- Hollan, J., Hutchins, E., and Kirsh, D. (2000). Distributed cognition: toward a new foundation for human-computer interaction research. *ACM Transactions on Computer-Human Interaction*, 7(2), 174–196.
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

- Kirsh, D., and Maglio, P. (1994). On distinguishing epistemic from pragmatic action. *Cognitive Science*, 18(4), 513–549.
- Klein, B., and Hoel, E. (2020). The emergence of informative higher scales in complex networks. *Complexity*, 2020, 8932526.
- Knorr Cetina, K. (1999). *Epistemic Cultures: How the Sciences Make Knowledge*. Harvard University Press.
- Latour, B. (1987). *Science in Action: How to Follow Scientists and Engineers Through Society*. Harvard University Press.
- Latour, B. (1993). *We Have Never Been Modern* (C. Porter, Trans.). Harvard University Press.
- Latour, B., and Woolgar, S. (1986). *Laboratory Life: The Construction of Scientific Facts*. Princeton University Press.
- Lave, J., and Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation*. Cambridge University Press.
- Levin, M. (2022). Technological approach to mind everywhere: an experimentally-grounded framework for understanding diverse bodies and minds. *Frontiers in Systems Neuroscience*, 16, 768201.
- Levin, M. (2025a). Ingressing minds: causal patterns beyond genetics and environment in natural, synthetic, and hybrid embodiments. *PsyArXiv* preprint, <https://doi.org/10.31234/osf.io/5g2xj>.
- Levin, M. (2025b). *Free Lunches: Model Systems for Studying the Agential Gifts from the Platonic Space* [lecture], <https://www.youtube.com/watch?v=K8BmMU1Tm-I>. The slide stating that the intelligence ratchet is a gift from mathematics is credited to Federico Pigozzi.
- Malafouris, L. (2013). *How Things Shape the Mind: A Theory of Material Engagement*. MIT Press.
- Netz, R. (1999). *The Shaping of Deduction in Greek Mathematics: A Study in Cognitive History*. Cambridge University Press.
- Noë, A. (2004). *Action in Perception*. MIT Press.
- Pigozzi, F., Goldstein, A., and Levin, M. (2025). Associative conditioning in gene regulatory network models increases integrative causal emergence. *Communications Biology*, 8, 1027.
- Pigozzi, F., and Levin, M. (2026). The causally emergent alignment hypothesis: causal emergence aligns with and predicts final reward in reinforcement learning agents. *arXiv* preprint arXiv:2605.06746.
- Polanyi, M. (1966). *The Tacit Dimension*. University of Chicago Press.
- Rotman, B. (2000). *Mathematics as Sign: Writing, Imagining, Counting*. Stanford University Press.
- Rupert, R. D. (2004). Challenges to the hypothesis of extended cognition. *Journal of Philosophy*, 101(8), 389–428.

- Shapin, S. (1994). *A Social History of Truth: Civility and Science in Seventeenth-Century England*. University of Chicago Press.
- Sterelny, K. (2010). Minds: extended or scaffolded? *Phenomenology and the Cognitive Sciences*, 9(4), 465–481.
- Sterelny, K. (2012). *The Evolved Apprentice: How Evolution Made Humans Unique*. MIT Press.
- Tomasello, M. (1999). *The Cultural Origins of Human Cognition*. Harvard University Press.
- Tomasello, M., Carpenter, M., Call, J., Behne, T., and Moll, H. (2005). Understanding and sharing intentions: the origins of cultural cognition. *Behavioral and Brain Sciences*, 28(5), 675–691.
- Vanchurin, V., Wolf, Y. I., Katsnelson, M. I., and Koonin, E. V. (2022). Toward a theory of evolution as multilevel learning. *Proceedings of the National Academy of Sciences*, 119(6), e2120037119.
- Varela, F. J., Thompson, E., and Rosch, E. (1991). *The Embodied Mind: Cognitive Science and Human Experience*. MIT Press.
- Walker, S. I., and Davies, P. C. W. (2013). The algorithmic origins of life. *Journal of the Royal Society Interface*, 10(79), 20120869.
- Watson, R. A., and Szathmáry, E. (2016). How can evolution learn? *Trends in Ecology and Evolution*, 31(2), 147–157.